

ALAN WRYA

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EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Science (BSc), Financial Mathematics Co-op | Computer Science Minor

Sept 2022 – June 2026

- **CGPA: 3.81** | Dean's List Scholar
- *Relevant Coursework:* Stochastic Calculus, Partial Differential Equations, Investment Analysis, Managerial Finance, Probability & Statistics, Linear Algebra, Differential Equations, Data Structures, Computer Science

EXPERIENCE

TD Bank

Jan 2025 – Aug 2025

Co-op, Direct Investing Journey Analyst

Toronto, ON

- Rebuilt 12+ monthly KPI dashboards in Excel using **VBA**, **LAMBDA** functions, dynamic named ranges, and reusable chart templates, reducing manual refresh time from 2-5 hours per week to under 5 minutes per dashboard.
- Engineered automated 24-month rolling performance charts with dynamic labels, $\pm 3\sigma$ control limits, and auto-scaling axes, enabling leadership to refresh reports on demand and standardize reporting across the team.
- Built client segmentation frameworks (Core Funded, Newly Funded, Reactivated, Attrited) and ran correlation and regression analysis on three years of monthly client data to surface drivers of retention, funding, and engagement.
- Led analytics tagging plans for new Direct Investing features: reviewed wireframes, scoped high-value events, presented to tagging and analytics teams, and coordinated risk sign-off across product, analytics, and 1B controls.
- Ran feature-launch lookback analyses on enhanced search adoption and mobile vs. desktop trading behaviour in Adobe Analytics and Excel to inform roadmap prioritization and executive-facing recommendations.
- Took on expanded analytics, stakeholder, and reporting responsibilities during the manager's parental leave.

Ted Rogers Sales & Trading (TRST)

Sept 2023 – Dec 2025

Quantitative Analyst

Toronto, ON

- Built a **Portfolio Optimization & Simulation tool** using historical data and mean-variance optimization to generate optimal Sharpe and minimum-volatility portfolios.
- Developed a **Value-at-Risk tool** using Monte Carlo simulations to assess potential portfolio losses from a distribution of outcomes.
- Adapted an interactive Black-Scholes option pricing calculator to compute call/put prices, implied volatility, and probabilities of expiring in-the-money.

R&R Utility Limited

Sept 2021 – Apr 2024

Project Management Assistant

Toronto, ON

- Supported the development of corporate operating budgets, project costing, and drawdown budgets across multi-discipline engineering and construction projects.
- Ran financial reconciliations on vendor invoices to resolve billing discrepancies and improve cost-tracking accuracy.

RESEARCH & PROJECTS

Cluster-Constrained Portfolio Optimization

2025 – 2026

Python, Scikit-learn

- Researched whether sector-clustering constraints improve mean-variance portfolio performance vs. plain MV on a 33-stock sector-stratified universe.
- Built an end-to-end backtesting pipeline: Ward hierarchical clustering → cluster-constrained MV optimization → benchmarking against Plain MV and ERC across 860 rolling 5-year windows.
- Showed cluster-constrained MV matches plain MV's out-of-sample Sharpe while materially lowering portfolio concentration, max drawdown, and 5% expected shortfall.
- Identified that plain MV behaves as an *error maximizer*, concentrating allocations in in-sample winners; cluster constraints mitigate this without sacrificing risk-adjusted returns.

Anomalies: Point-and-Click Trading Strategy Discovery Platform

2025 – Present

Python, DEAP, vectorbtpro, PySide6

- Core builder on a small team developing a point-and-click desktop application that lets users design, backtest, and stress-test systematic trading strategies across crypto and equity markets.
- Designed and built the backtesting engine, stress-testing pipeline, and evolutionary search system (DEAP) that discovers novel strategies by exploring combinations of technical-indicator signals and exit parameters.
- Built the desktop UI (PySide6) and a multi-source market data layer (Yahoo Finance, Binance, dYdX, Paradex, Hyperliquid) with local SQLite caching and Turso cloud sync.

SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn, Statsmodels, SQLAlchemy), SQL, MATLAB, R, Java, VBA

Quantitative & Finance: Mean-variance optimization, Monte Carlo simulation, Black-Scholes, Value-at-Risk, backtesting, time-series analysis, clustering, PCA

Tools: Adobe Analytics, Excel (LAMBDA, dynamic ranges, VBA macros), PowerPoint, Git, Jupyter

ACTIVITIES

Math Tutor (Self-Employed)

Sept 2022 – Present

Private High-School Mathematics Instruction

Toronto, ON

- One-on-one tutoring in high-school calculus, algebra, trigonometry, and geometry; built customized lesson plans tailored to each student's learning style and academic goals.